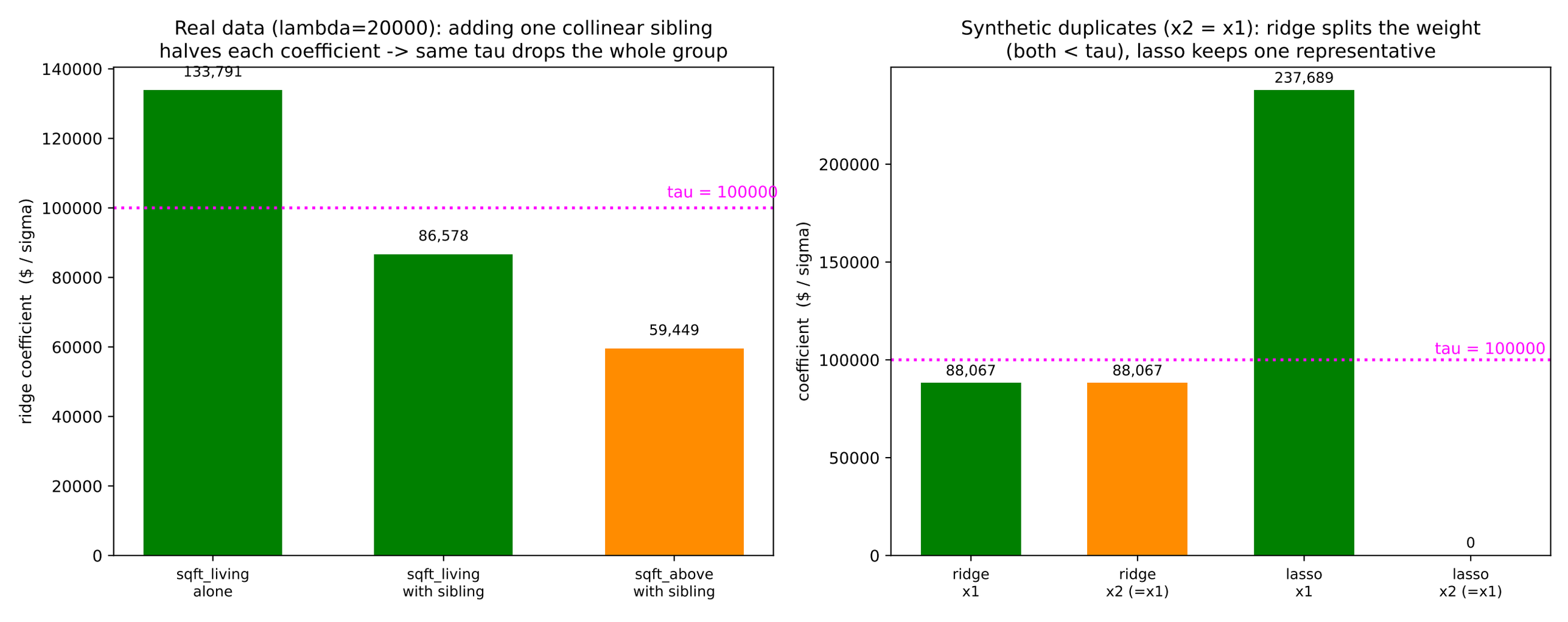
Real data (lambda=20000): adding one collinear sibling
halves each coefficient -> same tau drops the whole group
ridge coefficient ($ / sigma)
140000
120000
100000
80000
60000
40000
20000
0
133,791
86,578
59,449
tau = 100000
sqft_living
alone
sqft_living
with sibling
sqft_above
with sibling
Synthetic duplicates (x2 = x1): ridge splits the weight
(both < tau), lasso keeps one representative
coefficient ($ / sigma)
200000
150000
100000
50000
0
88,067
88,067
237,689
0
tau = 100000
ridge
x1
ridge
x2 (=x1)
lasso
x1
lasso
x2 (=x1)